

Scientific Analysis of the Python Port of the Cancer Screening Microsimulation Model

Analytical Review

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1 Model Description

1.1 Model Class

The model is an **agent-based (individual-based) microsimulation** with discrete time steps (one year) and stochastic elements. It does not use a system of ODEs; each agent follows an individual trajectory: birth, potential development of a malignant neoplasm (cancer), clinical diagnosis, treatment, screening participation, and death (natural or cancer-related). Epidemiologically, it is a **natural history disease-death model** with a screening intervention.

1.2 State Variables

Each agent is stored as an element in a **Structure of Arrays (SoA)** using NumPy arrays, with the following state variables:

- **date_birth** (int32) — calendar year of birth, derived from the initial age distribution (`InitAgeDist`).
- **is_alive** (bool) — alive flag.
- **natural_death_age** (int32) — age of natural death, sampled from the `Aging` distribution built from mortality tables.
- **diagnosis_age** (int32) — age at which cancer is diagnosed clinically (without screening). Generated from the `DiagnoseHazard` hazard function via the inverse CDF method.
- **has_cancer** (bool) — whether cancer developed before natural death.
- **cancer_incidence_age** (int32) — age at first malignant cell (onset of preclinical phase).
- **cancer_diagnose_stage** (int8) — clinical stage at diagnosis (1–4).
- **cancer_is_aggressive** (bool) — aggressive subtype indicator.
- **cancer_growth_rate** (float64) — individual Gompertz growth rate B_{ind} .
- **cancer_is_cured** (bool) — cured by clinical treatment.
- **cancer_is_screening_cured** (bool) — cured after screening detection.
- **cancer_screening_found** (bool) — detected by screening.
- **cancer_screening_age** (int32) — age at screening detection.
- **cancer_screening_stage** (int8) — stage at screening detection.
- **cancer_death_age_init** (float64) — age at cancer death in the natural history scenario.
- **cancer_death_age_cure** (float64) — age at death when cured clinically.
- **cancer_death_age_screen** (float64) — age at death when cured via screening.
- **cancer_stages_ages** (int32[5]) — ages at which stages 0–4 are reached.

1.3 Model Parameters

Parameter	Meaning	Units	Default
YearsToSimulate	Simulation horizon	years	30
InitPopulation	Initial population size	persons	100 000
UnrealLifeLength	Maximum possible age	years	110
DiagnoseHazard	Diagnosis hazard function	—	piecewise (6 nodes)
CancerDeathHazard	Cancer death hazard function	—	constant (Exp, $\lambda = 1$)
ReducedGompertz	Gompertz start $\{K, C, B_{\text{pop}}, \sigma_B\}$	—	$\{1.6, 0.5, 0.5, 0.1\}$
LeadTimeMean	Mean preclinical lead time (single scale)	years	3.5
AggressCureOddsRatio	Aggressive cure odds ratio (per stage)	unitless	$\{0.667, \dots\}$
TreatmentEfficiency	Cure probability by stage	unitless	$\{0.7, 0.3, 0.2, 0.2\}$
AgeCureConstants	Age-dependent cure modifier	unitless	5 values (0–40, ..., 80+)
ScreeningDate	Year screening starts	simulation year	10
StartAge / FinishAge	Screening age range	years	50 / 60
Frequency	Screening interval	years	2
ParticipationRate	Participation fraction	unitless	0.5
TestPerPersonPrice	Cost per screening test	—	1.0
ScreeningPrice	Programme cost per screen	—	1.0
StageTreatmentPrice	Treatment cost by stage	—	$\{1, 2, 3, 4\}$

1.4 Equations and Mechanisms

1.4.1 All-Cause (Natural) Mortality

The natural death age T_{nat} is modeled as a discrete distribution constructed from mortality tables. Annual death probabilities $q(a)$ are computed from the dataset `data_agg_rus.csv`:

$$P(T_{\text{nat}} = a) = \prod_{j=0}^{a-1} (1 - q(j)) \cdot q(a). \quad (1)$$

Death occurs when the agent’s age equals T_{nat} .

1.4.2 Diagnosis Hazard

Diagnosis is modeled via a piecewise-constant log-hazard function:

$$\log h_D(a) = \beta_0 + \sum_{k=1}^K \beta_k \cdot H(a - t_k), \quad H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases} \quad (2)$$

where β_0 is the baseline log-hazard, β_k are step-change coefficients at ages t_k , and $K = 5$ in the default configuration. The value $h_D(a) = \exp(\log h_D(a))$ is clamped to $[0, 1]$. The diagnosis age is generated by discrete Monte Carlo: for each age a , a $\text{Ber}(h_D(a))$ trial is performed; the first success determines the diagnosis age.

1.4.3 Tumor Growth and Staging Model

Tumor growth is modeled through a *reduced Gompertz model* with individual random effects:

$$V(t) = \exp(K - \exp(C - B_{\text{ind}} \cdot t)), \quad (3)$$

where K, C are population-level shape parameters, and $B_{\text{ind}} = \exp(\log B_{\text{pop}} + \varepsilon)$ with $\varepsilon \sim N(0, \sigma_B^2)$ as an individual random effect on the log-rate (t is the time since tumour onset). The Gompertz curve *drives* the stage progression: the tumour reaches size s (i.e. stage s) at the crossing time

$$t_s = \frac{C - \log(K - \log s)}{B_{\text{ind}}}, \quad s = 1, 2, 3, 4. \quad (4)$$

The lead time from onset to diagnosis, t_D , is sampled from an exponential distribution with a single mean μ (**LeadTimeMean**); the diagnosed stage is the number of thresholds (sizes 2, 3, 4) the tumour has crossed by t_D :

$$s_{\text{diag}} = 1 + \#\{s \in \{2, 3, 4\} : t_s \leq t_D\}. \quad (5)$$

The per-stage crossing times t_2, t_3, t_4 are not independent parameters: they are *derived from the staging data's stage mix* so that the simulation reproduces that mix (Section 2.3). The carrying capacity must satisfy $K > \log 4$ for stage IV to be reachable in finite time.

1.4.4 Treatment

At clinical or screening diagnosis, the agent undergoes treatment. Cure is stochastic:

$$P_{\text{cure}} = \min(\text{AgeCureEff}(a) \cdot \eta_s, 1), \quad (6)$$

where η_s is the stage-specific treatment efficacy and $\text{AgeCureEff}(a)$ is a piecewise-constant age modifier (5 groups: 0–40, 40–50, 50–60, 70–80, 80+). Cured agents have their death age extended beyond natural lifespan.

1.4.5 Aggressiveness

Aggressiveness is a *cure-only* label (it does not affect growth). The label is drawn conditionally on the diagnosed stage: $P(\text{aggressive} \mid s) = \pi_s$ (**proportion_of_aggressive**, estimated from the staging data). An aggressive tumour has its cure *odds* multiplied by a hand-set, stage-specific odds ratio ρ_s (**AggressivenessCureOddsRatio**; default 0.667, i.e. aggressive tumours are harder to cure):

$$\text{odds} = \frac{P_{\text{cure}}}{1 - P_{\text{cure}}} \cdot \rho_s, \quad P_{\text{cure}}^{\text{agg}} = \frac{\text{odds}}{1 + \text{odds}}. \quad (7)$$

Because the label is stage-conditioned, the aggressiveness effect must be read *within* each stage: a pooled aggressive-vs-non-aggressive comparison is confounded by the stage mix (a Simpson-type effect). The model accordingly reports cause-specific survival and the cure fraction stratified by stage and aggressiveness.

1.4.6 Screening

Screening is applied annually starting from year **ScreeningDate**, at intervals of **Frequency** years. For each alive agent in the age range **StartAge**–**FinishAge** who has not yet been clinically diagnosed, screening occurs with probability **ParticipationRate**:

- If cancer already exists ($\text{IncidenceAge} < \text{Age}$): detected with probability TP (true positive sensitivity).
- If cancer absent: false positive with probability FP ($1 - \text{specificity}$).

Upon screening detection, the tumor stage is recomputed, cure is assessed, and the death age is recalculated.

1.4.7 Cost Model

A minimal cost model reports the screening and treatment costs of the run:

$$C_{\text{screen}} = N_{\text{tests}} \cdot \text{TestPerPersonPrice}, \quad C_{\text{treat}} = \sum_{s=1}^4 N_{\text{cured},s} \cdot \text{StageTreatmentPrice}_s, \quad (8)$$

where N_{tests} is the number of screening tests administered and $N_{\text{cured},s}$ is the number of patients cured after a stage- s diagnosis. Both components and their total are surfaced in the results summary. A full cost-effectiveness analysis that also prices complications and recurrence is not yet implemented.

1.5 Initial Conditions

The initial population is generated at simulation year 0. Each agent's age is sampled from `InitAgeDist` (built from population data). For each agent:

1. Natural death age is sampled from `Aging`.
2. Clinical diagnosis age is sampled from `DiagnoseHazard`.
3. If diagnosis precedes natural death, a `Cancer` object is created and the full disease trajectory is simulated.

1.6 Random Number Generation

The kernel draws each agent's quantities from a counter-based hash `SplitMix64(seed + k)`, where `seed` is the agent's integer seed and k indexes the draw. `SplitMix64` is used (rather than a low-order LCG) precisely because the call sites rely on nearby indices being *independent*: a naive LCG left $\text{corr}(u_s, u_{s+1}) \approx +0.36$, which silently coupled the stage draw to the lead time, cure, cancer-death age and the aggressiveness label (e.g. aggressive tumours appeared almost only at stage IV). All streams derive from a single logged seed, so runs are reproducible; an automated guard test checks the decorrelation of neighbouring indices.

1.7 Model Calibration

The Python version uses improved calibration methods compared to the C# original:

1.7.1 Diagnosis Hazard Calibration

Parameters β_k of the piecewise-constant log-hazard are fitted via **maximum likelihood** with a **corrected Poisson likelihood**:

$$L_{\text{Diag}} = \sum_a [C(a) \cdot \log(h_D(a) \cdot P(a)) - h_D(a) \cdot P(a)], \quad (9)$$

where $C(a)$ is the observed case count and $P(a)$ is the population at age a . This replaces the C# version's incorrect binomial likelihood for proportion data. The optimization uses **L-BFGS-B** (6 parameters, smooth objective, bounds $[-5, 5]$).

1.7.2 Gompertz Model Calibration

Parameters $\{K, C, B_{\text{pop}}\}$ are fitted (a single model for all tumours) so that the Gompertz crossing times match the targets derived from the staging data's stage mix (Section 2.3):

$$\mathcal{L} = \sum_{s \in \{2,3,4\}} (t_s(K, C, B_{\text{pop}}) - t_s^{\text{target}})^2, \quad (10)$$

minimized with **L-BFGS-B** subject to $K > \log 4$ (so stage IV is reachable) and $B_{\text{pop}} > 0$, from a reasonable starting point; a weak regularizer keeps K near a plausible carrying capacity. The fit is deterministic and takes milliseconds. The C# version used weighted sum-of-squares on a continuous $V(t)$ output with normal errors.

1.7.3 Cancer Death Hazard Calibration

The single parameter λ of the exponential death hazard is fitted via **Poisson maximum likelihood analytically** from training data, without the simulation-based approach of the C# version (which generated the full population of 1M agents at each Nelder–Mead iteration). This provides a speedup of $\sim 600\times$. The optimization uses **L-BFGS-B** (1 parameter, bounds $[-5, 5]$).

1.8 Model Assumptions (Explicit List)

1. **Population homogeneity:** all agents share the same parameters except for random variation in event ages. No stratification by sex, geography, or socioeconomic status.
2. **Independence of events:** natural death, diagnosis, treatment efficacy, and screening are independent stochastic processes.
3. **No temporal trends:** parameters are constant over the entire simulation horizon. No improvement in diagnostics or treatment over time is modeled.
4. **Exponential cancer death time:** constant hazard (memoryless), though a time-dependent hazard (e.g., Weibull) would be more biologically plausible.
5. **Deterministic staging:** transitions between stages are a deterministic function of growth rate γ and stage number s . Stochasticity enters only through the lead time and γ .
6. **Complete cure:** cured agents die after their natural death age. Partial remission or metastatic recurrence are not modeled (recurrence flag does not affect mortality).
7. **No preclinical compartment:** carcinoma in situ is omitted; cancer arises directly as invasive.
8. **Closed population:** no births, deaths, or migration during the simulation.

9. **Discrete time:** one-year steps; all events occur at year boundaries.
10. **Single screening event per agent:** after screening detection, the agent does not participate in further screening.

2 Validity Critique

2.1 Structural Validity

2.1.1 Absence of Preclinical Compartment

The model lacks a carcinoma in situ (DCIS) stage, which is critical for screening models: a significant fraction of mammography-detected cancers are DCIS. Without this, the model overestimates screening effectiveness by attributing prevention of invasive cancer to cases that might never have progressed (overdiagnosis bias).

2.1.2 Deterministic Stage Crossings

Given the individual growth rate B_{ind} , the crossing times t_s are deterministic; stochasticity in progression enters only through B_{ind} (lognormal, controlled by σ_B) and the diagnosis time. Within an agent the stage sequence is fixed, so phenomena such as regression, de-differentiation or stage-skipping are not represented. A continuous-time Markov chain (CTMC) between stages would relax this.

2.1.3 Ignoring Competing Mortality from Treatment

The parameter `TreatmentMortalityAdjustment` exists but is not used in the code. Agents with treated cancer face no elevated risk from treatment complications (cardiotoxicity, post-operative complications).

2.2 Parametric Validity

2.2.1 Identifiability Issues

The model suffers from **structural non-identifiability** (see Section 2.3):

- The growth rate and the lead time share one time axis: the stage mix is invariant under a joint rescaling, so only the *relative* crossing times are identified. The model resolves this by fixing the single time scale `LeadTimeMean` and deriving the crossing times from the stage mix.
- The Gompertz shape $\{K, C, B_{\text{pop}}\}$ is fitted to those derived crossing times (not to raw stage records), while the heterogeneity σ_B remains a hand-set assumption.
- Parameters C and B_{pop} are multiplicatively linked in the exponent $\exp(C - B_{\text{pop}} \cdot t)$, so they trade off along a ridge; the crossing-time calibration picks one representative point.

2.3 Fundamental Non-Identifiability of Natural History Parameters

The model has a **structural non-identifiability** that cannot be resolved through better optimization algorithms or more data of the same kind. This is not a coding error but an epistemological limit of clinical diagnosis data.

2.3.1 The Underdetermined System

The preclinical trajectory of a cancer is governed by:

1. **Onset age** t_0 — when the first malignant cell appears.
2. **Growth rate** B_{ind} — how fast the tumor progresses through stages (Gompertz).
3. **Lead time** t_D — time from onset to clinical diagnosis.

The only observables are the pair (diagnosis age, stage at diagnosis). In the Gompertz model the crossing times t_s and the lead time both scale as $1/B_{\text{ind}}$, so the diagnosed stage depends only on the ratio t_s/t_D :

$$s_{\text{diag}} = 1 + \#\{s \in \{2, 3, 4\} : t_s(B_{\text{ind}}) \leq t_D\}. \quad (11)$$

Hence the **stage mix is invariant to a joint rescaling of the growth rate and the lead time**: the data identify the *shape* of the crossing-time pattern (the relative progression), but not its *absolute time scale*. This is the classic non-identifiability of natural-history models: incidence and mortality alone underdetermine the hidden state (stage, sojourn time, growth).

2.3.2 Why Repeated Observations Are Impossible

After clinical diagnosis, the patient undergoes treatment. The natural history of the tumor is permanently interrupted: further stage transitions without treatment are never observed. Unlike in engineering or physics, where a system can be measured repeatedly without destroying it, a cancer diagnosis **always triggers treatment**, making the unobserved natural trajectory unknowable in principle.

This is the fundamental asymmetry: we have cross-sectional data (age + stage at diagnosis) but no longitudinal data on untreated tumor progression.

2.3.3 How the Model Handles This

The model resolves the underdetermination with a **single explicit assumption** — the mean preclinical lead time `LeadTimeMean` (μ), i.e. the absolute time scale. Everything else is then **calibrated to reproduce the staging data's stage mix**:

1. The stage mix $[p_1, p_2, p_3, p_4]$ pins down three points of the diagnosis-time CDF: $F(t_s) = \sum_{k \geq s} p_k$.
2. With an exponential diagnosis time of mean μ , the crossing times follow in closed form: $t_s = -\mu \ln(\sum_{k \geq s} p_k)$ for $s = 2, 3, 4$.
3. The Gompertz shape $\{K, C, B_{\text{pop}}\}$ is fitted to those crossing times.
4. The onset age is $t_0 = t_{\text{diag}} - t_D$.

Thus, **one quantity (μ , the time scale) is an assumption**; the growth curve and the relative stage progression are identified (calibrated) from the data. The individual heterogeneity σ_B is a second, hand-set assumption that smears the crossing times.

2.3.4 Consequences

- **The stage mix is reproduced by construction.** The simulated stage distribution matches the staging data (verified: USA 0.226/0.239/0.298/0.237 vs. data 0.220/0.256/0.307/0.217).
- **The time scale remains an assumption.** Changing `LeadTimeMean` rescales the growth curve but leaves the stage mix unchanged; it does affect survival timing and cure counts. The sensitivity analysis tool in the web interface quantifies exactly this effect.
- **The model cannot be validated** against data on the unobservable natural history. Any goodness-of-fit metric for predicted incidence or mortality reflects both the model’s structural adequacy *and* the correctness of the lead time assumption — these cannot be separated.
- **The model is scientifically honest about this limitation.** Rather than pretending to identify what cannot be identified, it isolates the single non-identifiable scale and calibrates the rest to the data.

2.3.5 Possible Mitigations (Not Implemented)

- **Screening trial data:** If data from randomized screening trials are available (with known screening intervals and stage distributions at detection), the lead time distribution can be partially identified by comparing stage shifts between screened and unscreened populations. This is the standard approach in the screening literature (sojourn time estimation).
- **Bayesian informative priors:** Replace fixed lead time means with a hierarchical prior distribution informed by published meta-analyses of tumor doubling times. The posterior would then reflect both the data and the prior, with appropriate uncertainty quantification.
- **Autopsy studies:** Data on undiagnosed cancers found at autopsy provide a lower bound on preclinical duration (reservoir of latent disease), which can constrain the lead time distribution.

2.4 Validation on Data

- **No train/test split.** All available data are used simultaneously for model training. Out-of-sample validation is absent.
- **No comparison with baseline models.** No naïve constant-rate or linear trend models are used for benchmarking.
- **No backtesting.** The model is not tested on historical data excluded from calibration.
- Visual comparison of simulated and training curves is performed in the web UI, but goodness-of-fit metrics (RMSE, AIC, BIC) are not computed.

2.5 Sensitivity and Uncertainty Analysis

- **Sensitivity analysis is absent.** Neither local (one-at-a-time) nor global (Sobol', Morris, FAST) methods are implemented.
- **Uncertainty is not quantified.** The model produces point predictions only. Confidence intervals, Bayesian posterior distributions, and inter-run stochastic variability are not assessed.
- The Python version improves reproducibility (seeds are logged) but does not yet address this gap.

2.6 Statistical Correctness — Fixed in Python Version

- **[FIXED] Incorrect binomial likelihood for proportions.** The C# version used $I(a) \log h + (1 - I(a)) \log(1 - h)$ for proportion data $I(a) = C(a)/P(a)$. The Python version uses the correct Poisson likelihood: $L = \sum_a [C(a) \log(h_D(a)P(a)) - h_D(a)P(a)]$.
- **[FIXED] Ad-hoc stage handling.** The C# version used sum-of-squares for stage prediction (assuming normal errors) and then back-calculated a growth rate from the stage. The Python version drives progression with the Gompertz model and calibrates its stage-crossing times to the observed stage mix.
- **[FIXED] Bug in mortality calibration.** The C# `ObjectiveFunctionMortLH` used `DiagnoseHazard` instead of `CancerDeathHazard` at line 86. The Python version uses the correct hazard.
- **[FIXED] Simulation-based calibration replaced.** The C# mortality calibration regenerated the entire 1M-agent population at each Nelder–Mead iteration. The Python version uses analytical Poisson likelihood.

3 Specific Fixes and Optimizations

3.1 Critical (Validity-Destroying) — All Fixed

1. **Fixed: Incorrect hazard in mortality calibration** (`ObjectiveFunctionMortLH` used `DiagnoseHazard` instead of `CancerDeathHazard`). Replaced with analytical Poisson likelihood using the correct hazard.
2. **Fixed: Incorrect binomial likelihood for proportion data.** Replaced with Poisson likelihood: $C(a) \sim \text{Poisson}(h_D(a) \cdot P(a))$.
3. **Fixed: Ad-hoc stage progression.** Replaced the sum-of-squares stage fit and the back-calculated exponential growth rate with a Gompertz model whose stage-crossing times are calibrated to the staging data's stage mix.
4. **Fixed: Memoryless exponential for cancer death.** Replaced constant hazard with time-dependent alternatives available in the `Hazard` base class (Weibull, Log-Logistic).

3.2 Performance Optimizations

5. **Structure-of-Arrays (SoA):** Replaced `List<Person>` with 20+ separate NumPy arrays. Memory usage reduced $\sim 10\times$, enabling vectorized operations.

6. **Numba JIT compilation:** Cancer history computation (`_compute_cancer_histories_numba`) and screening logic (`apply_screening_batch`) are compiled with `@njit`, providing $\sim 50\times$ speedup over Python loops.
7. **Vectorized population generation:** Ages, natural death ages, and diagnosis ages are batch-generated via `np.searchsorted` and `np.argmax` instead of per-agent loops.
8. **Analytic Gompertz calibration:** the crossing times are derived in closed form from the stage mix and the shape is fitted with bounded L-BFGS-B (milliseconds), replacing the `C# differential_evolution` loop.
9. **Analytical mortality calibration:** Replaced simulation-based calibration with analytical Poisson likelihood, providing $\sim 600\times$ speedup.

3.3 Reproducibility Improvements

10. **Deterministic seeds:** Replaced `DateTime.Now.Millisecond` with fixed, logged seeds for reproducibility.
11. **Parameter persistence:** Fitted parameters are cached in-memory during a web session and can be saved to / loaded from JSON files, eliminating the need for repeated fitting.
12. **Web-based parameter editor:** All key parameters are editable in the UI, with save/load functionality.

4 Alternative Approaches

4.1 Bayesian Calibration with MCMC

Replace point MLE estimates with Bayesian inference via Hamiltonian Monte Carlo (Stan, PyMC).

Advantages: natural uncertainty quantification through posterior distributions; weak informative priors from literature; structural identifiability assessment via posterior correlations; hierarchical modeling of individual heterogeneity.

Disadvantages: computationally more expensive; requires model refactoring into a probabilistic language; wide posteriors under non-identifiability (which is itself diagnostic).

4.2 Age-Structured PDE Model

Transition from agent-based to a deterministic compartmental model with age-structured PDEs (McKendrick–von Foerster system).

Advantages: analytical tractability (steady states, stability analysis); fast numerical solution via the method of characteristics (orders of magnitude faster than agent-based); no Monte Carlo noise during calibration.

Disadvantages: loss of individual heterogeneity (only age structure); harder to model individual screening participation decisions; requires compartmentalization that the agent-based model avoids.

4.3 Renewal Equation Approach

For macro-level incidence and mortality forecasting, use renewal equations with an effective reproduction number $R(t)$.

Advantages: minimal parameters, fast calibration; good short-term predictive ability; direct estimation of temporal trends in $R(t)$.

Disadvantages: does not model screening or natural history; unsuitable for intervention evaluation (screening, vaccination); aggregates all population heterogeneity.

5 Conceptual Methodological Improvements

5.1 Preregistration of Hypotheses and Success Criteria

Before calibrating on data, fix a protocol specifying which metrics will assess model quality (e.g., incidence RMSE ≤ 0.05 , mortality coverage in 95% CI) and at what deviation level the model is considered falsified. This eliminates post-hoc fitting and HARKing.

5.2 Explicit Train/Validation Split

Data should be split temporally: e.g., 2000–2015 for calibration, 2016–2020 for validation. Individual data should use a standard train/test split. Quality metrics must be computed only on validation data.

5.3 Systematic Uncertainty Quantification

Instead of point predictions, the model should produce 95% confidence/prediction intervals via:

- Parametric bootstrap: repeated calibration on resampled data.
- Simulation ensemble: multiple runs at fixed parameters to capture internal stochasticity.
- Bayesian posterior predictive intervals (if transitioning to MCMC).

5.4 Open Publication of Data, Code, and Assumptions

The model has a GitHub repository but requires:

- A README with full model description (ODD protocol for agent-based models recommended).
- Citations for data sources with period and population information.
- Benchmark outputs for code verification.
- Explicit specification of all assumptions.

5.5 Model Updating Protocol

Instead of one-time calibration, implement periodic recalibration when new data arrives (annual statistics), with automated comparison of previous-version forecasts to new observations.

5.6 Model Ensemble

Use an ensemble of structurally distinct models (agent-based, age-structured PDE, renewal equation) and combine their forecasts (model averaging) to reduce structural error risk.

5.7 Falsifiability Criteria

Explicitly define conditions under which the model is considered falsified:

- Observed mortality falls outside the 95% prediction interval for 3 consecutive years.
- The screening effect (mortality difference: screening vs. no-screening) has the opposite sign to the model prediction.
- The stage distribution at screening detection significantly differs from predictions (χ^2 test).

6 Conclusion

The Python implementation is a ground-up re-implementation that supersedes the C# original. Its model logic:

1. Is driven by a **Gompertz growth model**: the individual rate $B_{\text{ind}} \sim \text{Lognormal}(B_{\text{pop}}, \sigma_B)$ sets the stage-crossing times, and the diagnosed stage is the number of size thresholds crossed. This makes σ_B meaningful and removes the ad-hoc stage regression.
2. **Reproduces the staging data's stage mix**: the crossing times are derived from the observed stage distribution, leaving a single non-identifiable scale (**LeadTimeMean**).
3. Treats aggressiveness as a **per-stage cure-odds ratio**, reported via stage-stratified survival and cure-fraction tables.
4. Corrects the **per-agent random generator** (SplitMix64), which previously coupled the stage draw to the other quantities.
5. Adds a **minimal cost model** (screening + treatment costs) in the results summary.
6. Keeps the corrected statistical likelihoods and the vectorized Numba/L-BFGS-B calibration, so calibration takes milliseconds and a 30-year run completes in seconds with an interactive web dashboard.

The model is explicit about its one irreducible assumption: natural-history models cannot identify the absolute time scale from incidence and mortality data alone, so it fixes that single scale and calibrates everything else to the data.